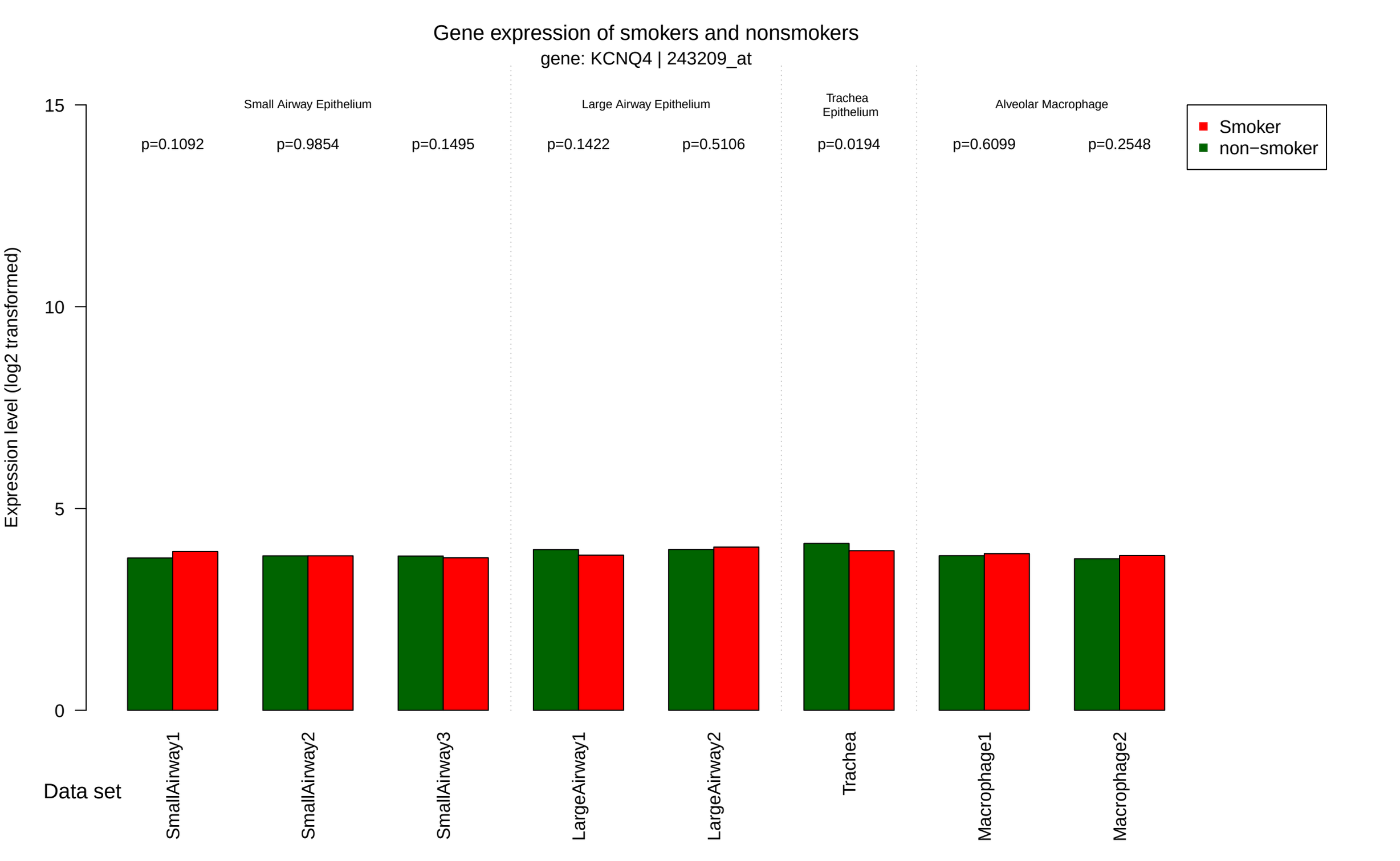
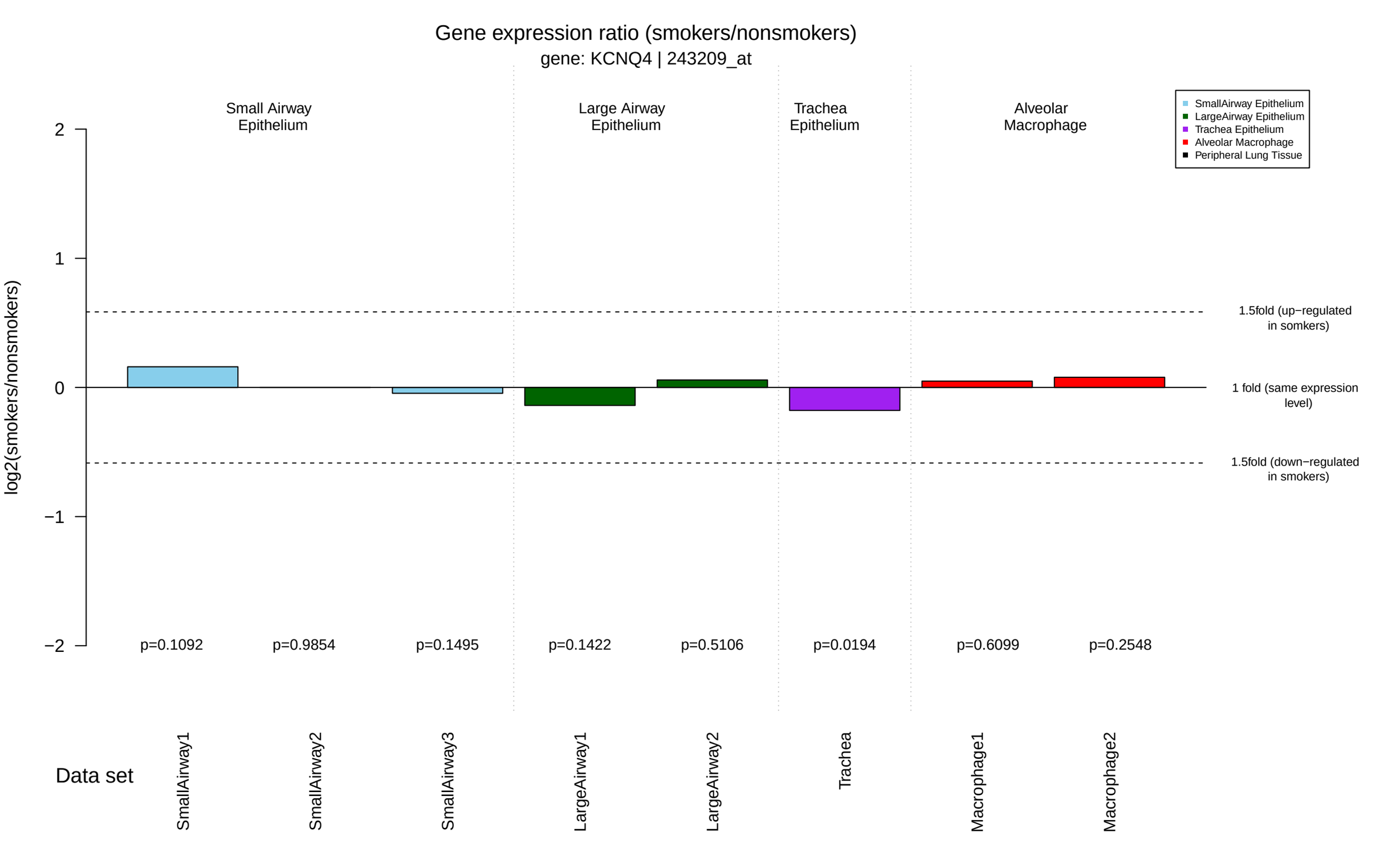
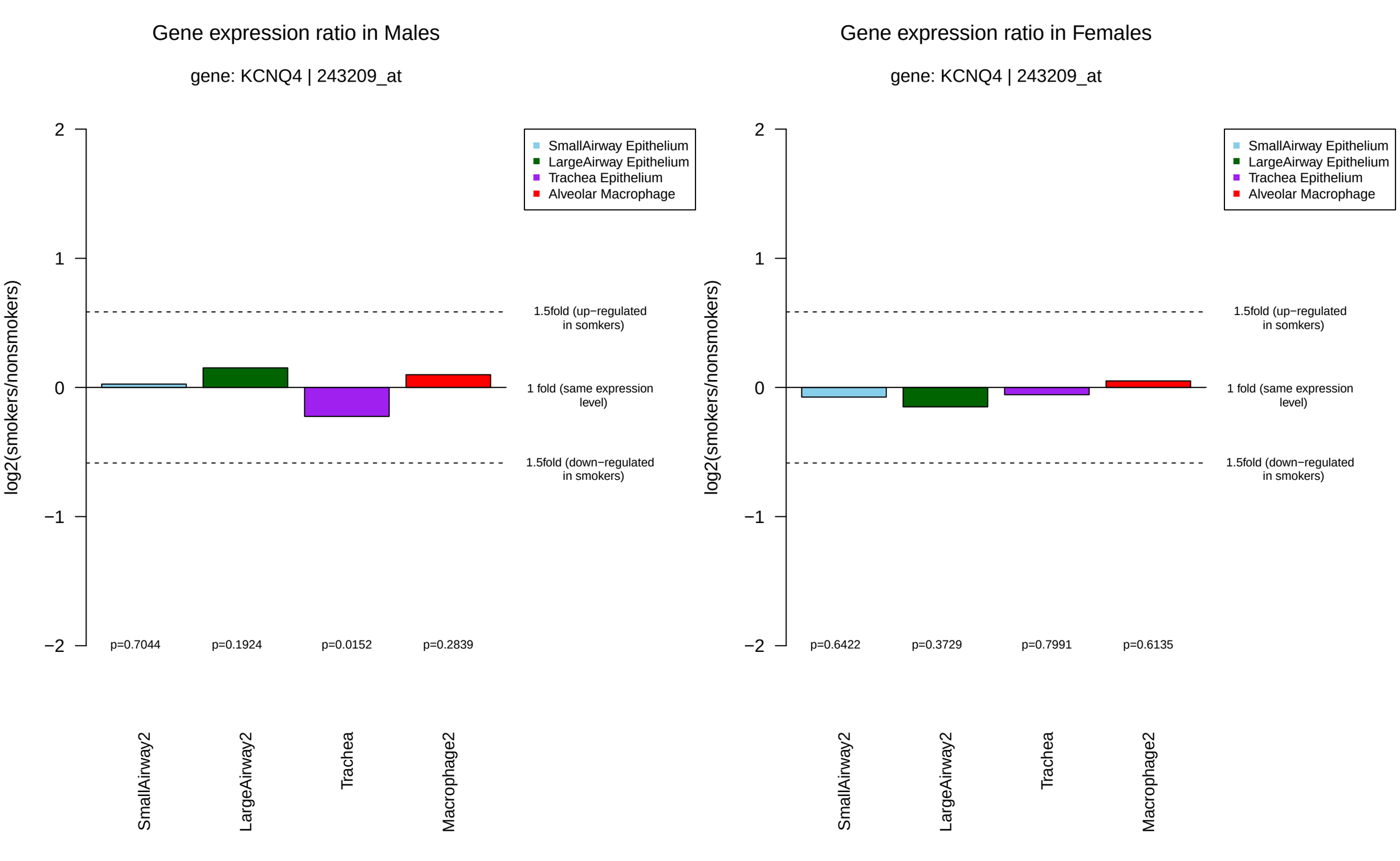
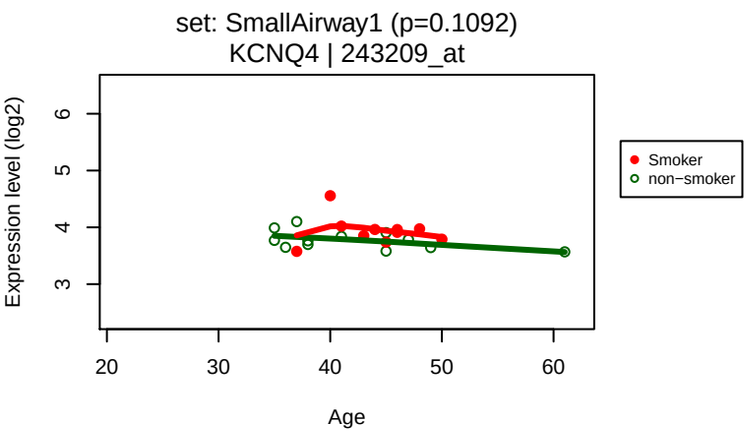
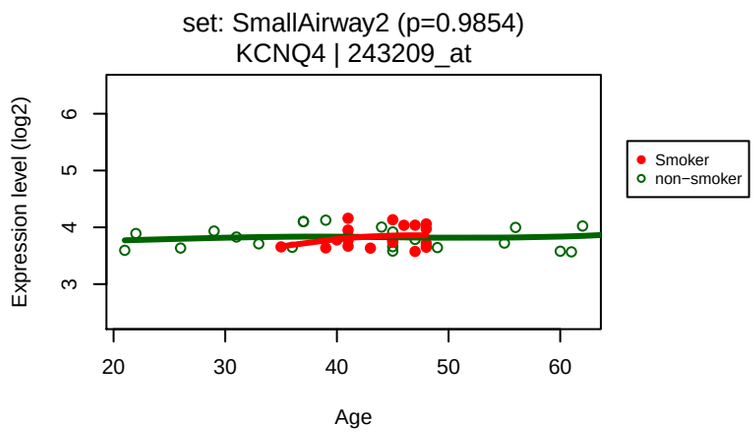
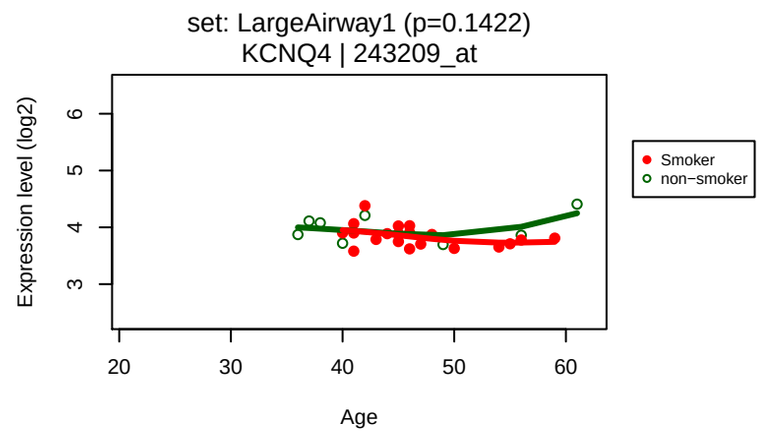

Gene expression of smokers and nonsmokers
gene: KCNQ4 | 243209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1092
p=0.9854
p=0.1495
p=0.1422
p=0.5106
p=0.0194
p=0.6099
p=0.2548
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

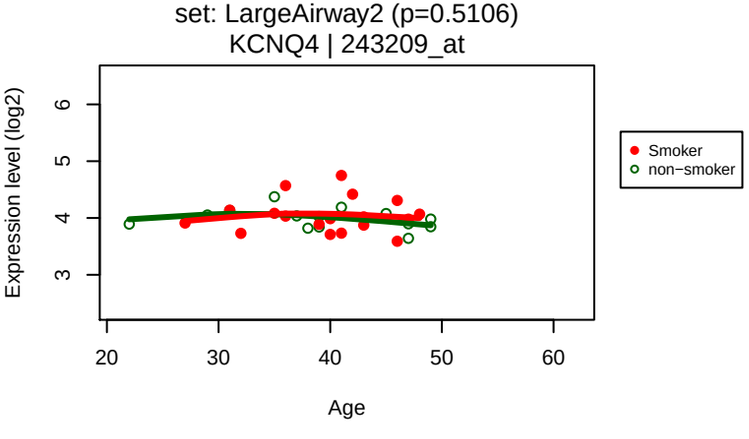

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ4 | 243209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1092
p=0.9854
p=0.1495
p=0.1422
p=0.5106
p=0.0194
p=0.6099
p=0.2548
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

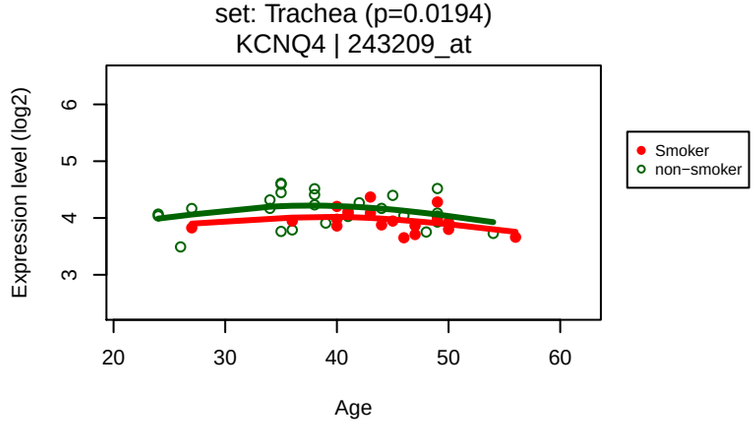

Gene expression ratio in Males
gene: KCNQ4 | 243209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7044
p=0.1924
p=0.0152
p=0.2839
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ4 | 243209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6422
p=0.3729
p=0.7991
p=0.6135
SmallAirway2
LargeAirway2
Trachea
Macrophage2

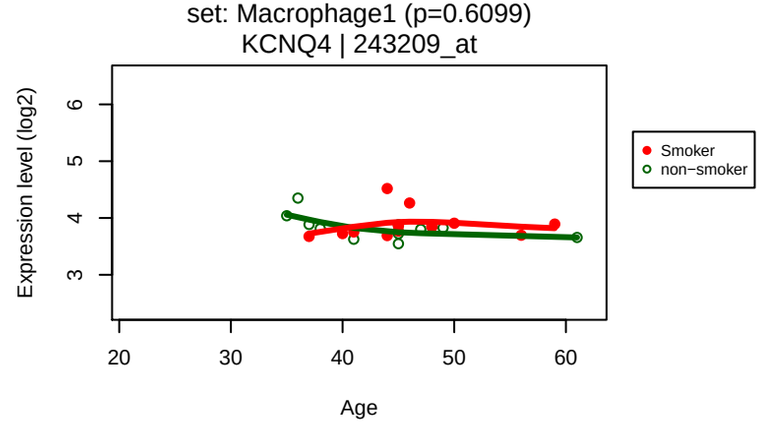

set: SmallAirway1 (p=0.1092)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker

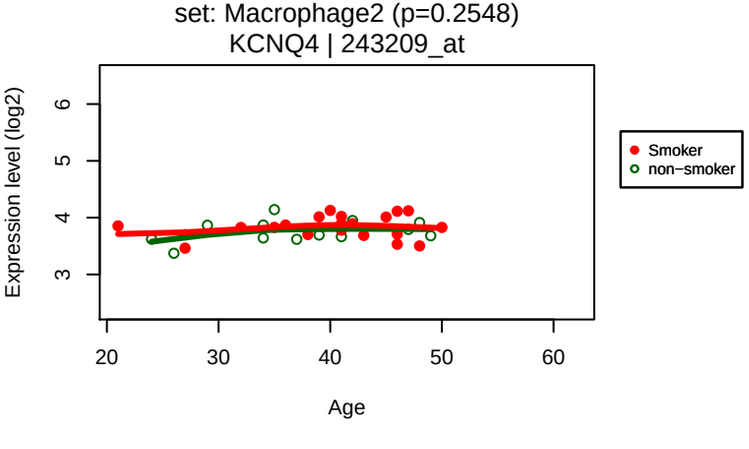

set: SmallAirway2 (p=0.9854)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker

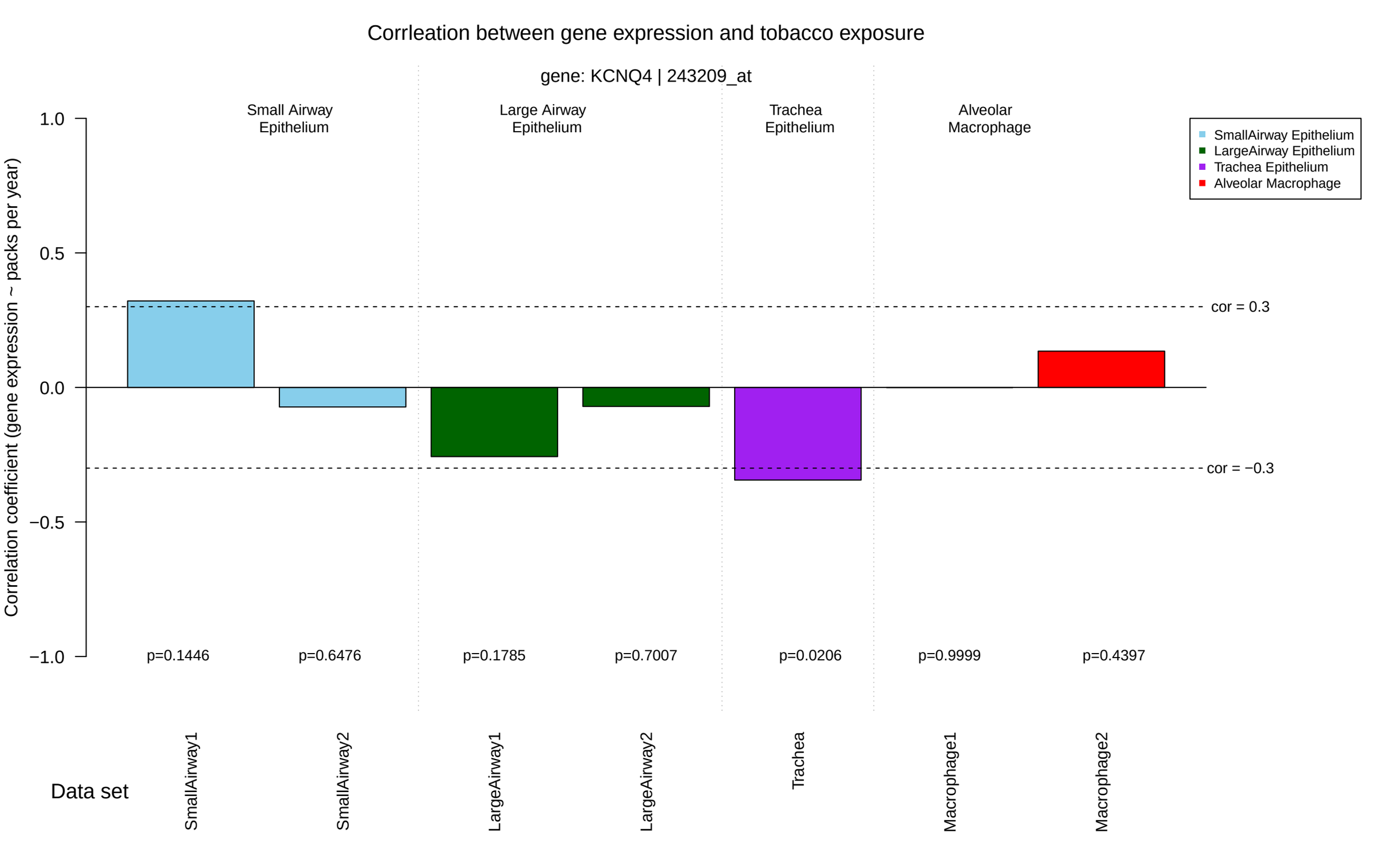

set: LargeAirway1 (p=0.1422)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5106)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0194)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6099)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2548)
KCNQ4 | 243209_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ4 | 243209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1446
p=0.6476
p=0.1785
p=0.7007
p=0.0206
p=0.9999
p=0.4397
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2